

Active Inference ~ Parr, Pezzulo, Friston ~ Chapter 8

~ BookStream #002.02

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chapter 8 is called active inference and continuous time begins with that Timeless quote everything flows nothing stands still what would you say about chapter eight all right so this chapter probably is my most favorite chapter in the book uh because of my own personal interests and uh I don't know the process materialism and so on but uh yeah so chapter seven um it acts as a really good starting point for anyone who wants to develop the discrete time situations and to model the screen time situations uh inactive within active inference framework but in chapter 8 we kind of get into get to a Model A bit more interesting or let's say more involving situations uh and they're not necessarily uh kind of toy examples we saw at least at the beginning of chapter seven uh so uh obviously as the title suggests this chapter deals with the continuous time situation so uh in the in that case we'll need to and maybe at this point um refresh our memory about what continuous time situation involves by reading the relevant Parts reading relevant parts of chapter four so uh yeah in chapter four we saw that the generative model for continuous time situation uh derives derives from uh the uh stochastic calculus in terms of putting the whole process into two elements uh two stochastic two elements of stochastic equations one of which is the actual State the condition of actual States or the behavior of the actual States and the other one is the randomness that we need to account for in each real-time continuous time situations so that's what we get here in equation eight a eight point one and then building up from that equation uh we um it generalizes that equation to involve uh I mean the fun the functionals of G and F instead of just the [Music] the single valued functions of G and F so uh then we get to put that into uh the situation that can be used for describing the behavior dynamical systems which is a very well-known uh situation to use these kinds of stochastic equations and it's widely studied how those those kinds of Dynamics can be characterized especially in recent Bayesian mechanics paper by Dalton sectivitavel and others so and then it gets to some more specific examples such as a

lockable terror Dynamics and synchronicity and so on in order to show how these kinds of Dynamics can be elaborated upon and can be generalized to and enables them to characterize a more complex situations so uh yeah that's a really short and brief overview of the whole uh chapter maybe uh we can talk about a bit more details as we go through right well said well I'm sure for another day the philosophical implications of eight seven and eight and high road and low road all these other parts of the textbook great topics um I I agree I would see chapter 8 as demonstrating continuity with some classical continuous time modeling motifs from a few different areas of dynamical Systems Science which is applied in like many many many fields but these are some classical examples so figure 8.1 goes a little bit more into depth or at least into more formalism detail about exactly what we saw in chapter five with the spinal reflex arc with the proprioceptive data coming in and then a differential being calculated with the set point which reflects a descending prediction from a decision-making layer and that can be viewed as this kind of mechanics that plays out in a phase space in continuous time like a spring moving around with with someone making a certain path with an attractor and a spring being dragged around something in that area box 8.1 goes into a very fascinating topic do you want to describe it uh well uh it's uh maybe one of the most um thought-provoking uh section pages of the whole book and if I remember correctly uh in all of the cohorts this particular box I mean gives always gives rise to lots of questions uh because of some of the interesting and uh at least initially counter-intuitive claims here but I don't want to spoil it but as a kind of spoiler alert it kind of gets to a really interesting but uh alas very brief discussion about the comparing these terms Precision attention and sensory the attenuation and the relation and similarities and difference between these two these three terms and how each understanding each of them is essential to understanding uh the other ones but as I said it's a really interesting topic which gives rise to lots of discussions and I believe it's one of those topics that that's worth looking a bit more uh looking into in some other literature as well great well set what a cliffhanger next they go to a classic Model family called lockabletera these Dynamics inherit from characterizations of Predator prey Dynamics in ecology so it's kind of a classical ecology model shown in figure 8.2 on the top it's actually the ecosystem model plants herbivores and carnivores which follow different kinds of oscillatory Trends in continuous time and so that also has enabled it to be applied for other so-called winnerless competitions and that relates to topics like neural Darwinism and also neurodynamics where things have kind of oscillatory relationships with each other which are being modeled as a continuous time underlying process with a lot of measurement noise and discretization through space and time

those are the kinds of algorithms that SPM explores more and there's lack of Volterra and a lot of other dynamical systems theory in SPM so active inference kind of adds action and more to what was laid out from a pure dynamical systems theory in SPM here it really is just showing the ecology example and how you can project if you have three different species you can think about that motion in a cube or tetrahedron and then you could project onto kind of like uh looking at a lower dimensional manifold relating just two of the three species and that advances this kind of oscillatory but also moving Behavior that gets connected in figure 8.3 to neurobiology what would you say about this uh okay so here in figure 8.3 uh we see some applications of um a lock of Altera Dynamics uh so uh the left column uh here uh represents um what happens in I mean in eye blinking eye blink conditioning so uh of course here we need to account for uh I mean the expected States of uh of the of the sequel of the sequences of events that happens in in the eye blinking so uh the upper upper left figure shows the expectations uh in terms of time and then uh the parallel right hand side equation sorry right hand side figures uh shows the uh the lock of Altera system that that is applied um in the handwriting situation so as we can see although uh the uh I mean mathematical technology is the same or at least the modeling technology is the same the outcome of each situation uh varies drastically in two distinct the two distinct neurobiological Behavior not neurobiological but biological Behavior Uh so yeah we can see how the same modeling framework can give rise to different uh outcomes uh based on what parameters is needs to be optimized what parameters are selected for the modeling and so on so I believe it's a quite uh interesting example to compare handwriting and the blinking together and how those can be compared to each other using the last laughable Terror Dynamics great thank you box 8.2 gives a variance on the learning here presented with the formalism for continuous models kind of a technical aside section 8.4 is about generalized synchrony so figure 8.4 is going to visualize one of the classic dynamical systems which is the Lorenz attractor so what would you say about this figure okay so this section is uh truly interesting because uh when one thinks of active inference probably the first situations that comes to mind is uh the situations in which we have a quite well-defined uh probability distributions for different parameters but as we can see here in section 8.4 actually some of the formalism of active inference can be successfully used to characterize even chaotic systems and protect and in particular the way in which two chaotic systems can be synchronized with each other so this is a classic example of a chaotic Lorenz system uh and um it derives from it draws upon uh from some of a professor person's earlier work on Birdsong synchrony and as a side note any literature before 2016 is considered

earlier history in active inference literature because it evolves uh quite rapidly so uh yeah this kind of synchrony between two chaotic systems uh can be interpreted as providing evidence or even let's say a way to model a kind of primitive theory of mind uh in the sense that how exactly can we understand uh or can two agents can trace each other's trajectories uh without um any I mean engaging in any direct exchange of observations between their internal and external States so uh yeah that's a really good example and I believe one of the most interesting examples of how active inference can even account for these kinds of uh Behavior so and the rest of the section goes into the details of how this kind of synchrony between um multi multi-scale uh Birds multi-scale Lorenz systems can happen um and how can we formulate it mathematically in terms of continuous time active inference awesome and there's been more recent work on Markov blankets and stochastic chaos but the bird example is a classic 8.5 goes into hybrid discrete and continuous models so this could be kind of like a in-between chapter of seven and eight but now that we've been introduced to the pure form of discrete in the pure form of continuous models here shown the dot composability extends to so-called Hybrid models where here the lower level visually is using the continuous time formalism and the higher level is describing the little line added here the discrete time formalism and this was the similar structure described by the authors of the paper octave inference does not contradict folk psychology where they describe this lower level as motor active inference which was closely allied with the spinal Arc reflex shown above and then this higher level they call decision active inference because in that case it was referring to a discrete decision and so they used that kind of basic motif of continuous activity or continuous time modeling at the more peripheral aspects of a cognitive entity and like Ali said more discretization and hybridization as well at higher levels of the cognitive modeling and that type of an architecture here instead of describing who wants the ice cream cone I believe here it's gonna be a mixed or hybrid model that is going to call back the isocade system where there's a fixed point that is able to be moved as a set point and then there's a continuous time isocade that pursues the new fixed point so that's analogous to a new set point or fixed point being specified from the top down muscle command about a new location for a muscle followed by movement towards it this is a muscular activity that is realizing that but but not in the elbow coming away from the hot stove this is about the eye cicading to an epistemic foraging location specified by top down hierarchical systems 8.3 describes little technical aside on mixture of gaussian gaussian mixture models kind of a technical modeling note an 8.6 closes it says it's a huge topic and much has been left out and so they list in table 8.1 key

advances in continuous time models and those areas are synthetic bird song ocular motor delays conditioned reflexes smooth Pursuit eye movements psychosis illusions cicades action observation attention Hybrid models and self-organization and that's chapter 8. what else would you say and also what would you kind of lead someone to in the philosophical implications of eight because it sounds kind of cool uh okay so uh well the case of continuous time active inference uh I think it leads to a really interesting questions uh both in terms of uh philosophical questions and also uh a more practical modeling questions about uh what parameters needs to be accounted for and so on uh and as I said I believe it's a more more interesting way of um if not interesting but but at least more involved way of doing active inference modeling but uh one thing that uh one of the philosophical questions that um have explored uh in our paper is how the processes of I mean ontological processes can uh philosophically described using fpp assertions in terms of their intraction with the environment [Applause] in which they co-constitute themselves and we don't necessarily distinguish between uh between the internal and the external state so one obvious example of this is that generalized synchrony example that we saw in this chapter in which we don't necessarily distinguish between which of the birds act as the agents and which one is the um environment or the vice versa so these kind of co-constitution of the environment and the agent which gives rise to the partitioning of State space through a markup blanket is uh one of the interesting uh philosophical points that I think needs to be elaborated a bit more using and some of the recent advances in philosophy such as the two the tools that's been developed in new materialism school or some other philosophical approach approaches but yeah these kinds of uh what exactly gives size gives rise to emergence what is the ontological status of emergent properties and so on are some of the burning questions for many philosophers today and I believe active inference in particularly continuous time active inference provides a clear precise mathematical formalism uh even if not to answer these questions but at least to explore it in a more rigorous and uh practical way and also practical and attractive attractable way so this is the area that I believe uh philosophy and science uh are beautifully intertwined uh into a coherent view of not only the phenomenon of interest but even about the whole world wow wow pretty cool yeah a lot to say about that topic after completing chapter seven and eight you've seen the kind of two major branches or two major motifs of just one kind of modeling but these kind of models have so many different forms that that's why it's such a Hands-On process to specify the generative model in chapter six and fit it with data in chapter nine those are all what's required and that's kind of the last mile of where these discussions about General

motifs gets you but also playing with these pedagogical models can be really helpful because it will help you understand the basic patterns and relationships and start to see see different patterns in the graphical models and know from there what levels of technical processes can be kind of coarse grained over